

Roll No. _____

23/3107
B.C.A. (Fifth Semester)
Examination, 2023
Third Paper
(Computer Networks)

Time : 3:00 Hours / *Maximum Marks : 75*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) What are the basic elements/ components of a Data Communication System? Explain with diagram. 8
- (b) Write a brief note on Distributed Processing Systems. 7

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2. What do you understand by the term 'Network Topology'? Compare the relative advantages and disadvantages of different types of Topology with diagram. 15
3. Differentiate between: 5+5+5
 - (i) OSI Reference Model and TCP/IP Model.
 - (ii) Modems and Network Interface Card
 - (iii) Simplex and Full duplex Transmission Mode.
4. Write a short note on : 5+5+5
 - (i) Multiplexing in Networking
 - (ii) Synchronous and Asynchronous Protocols
 - (iii) Connection-Oriented and Connection-less services in a Network.

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5. What is the range of services offered under ISDN? Give an account of the layers in ISDN. What do you understand by B-ISDN? Explain. 15
6. Explain the role of the following in a Computer Network : 3×5=15
- (i) Repeaters
 - (ii) Bridges
 - (iii) Gateways
7. Explain how the following techniques are implemented in a Network : 15
- (i) Congestion control
 - (ii) Internet working
 - (iii) Authentication

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8. What are the functions and services offered in : 15

- (i) Transport Layer
- (ii) Sessions Layer

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